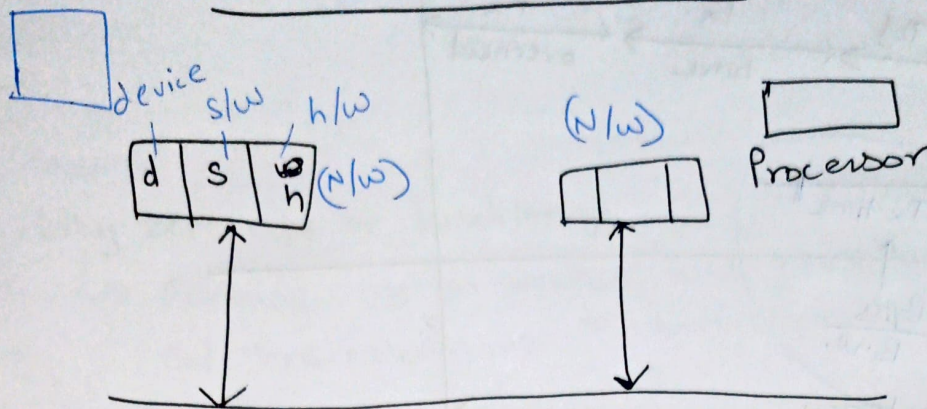


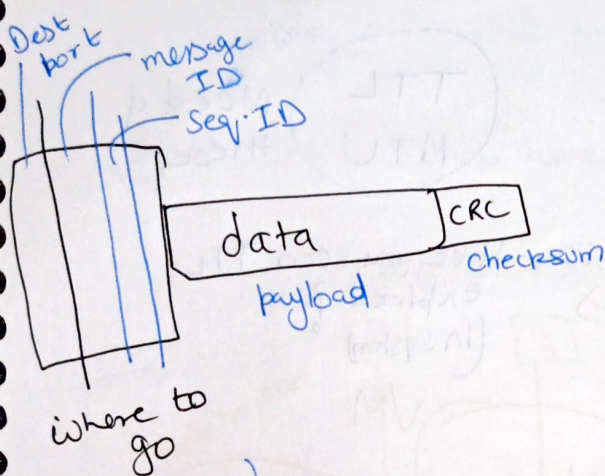
13-oct-25

Processor

Interconnection Networks



This is how N/w talk to each other via messages



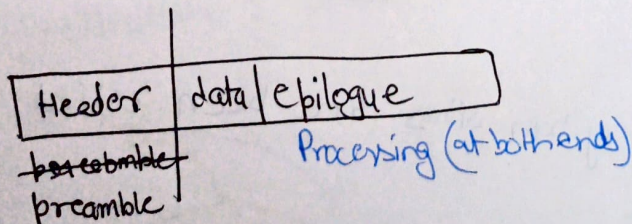
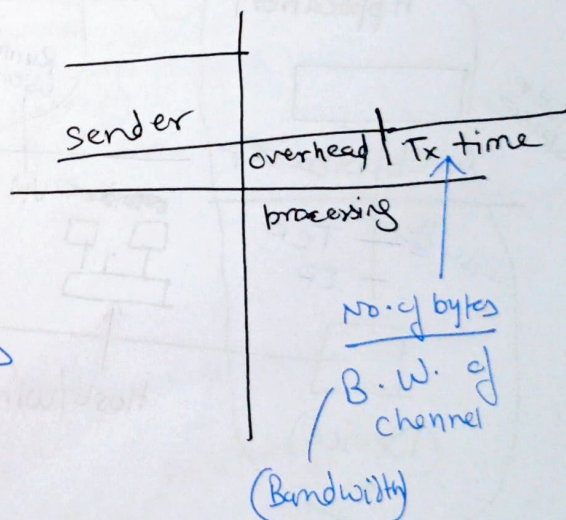
Data you are interested in sending

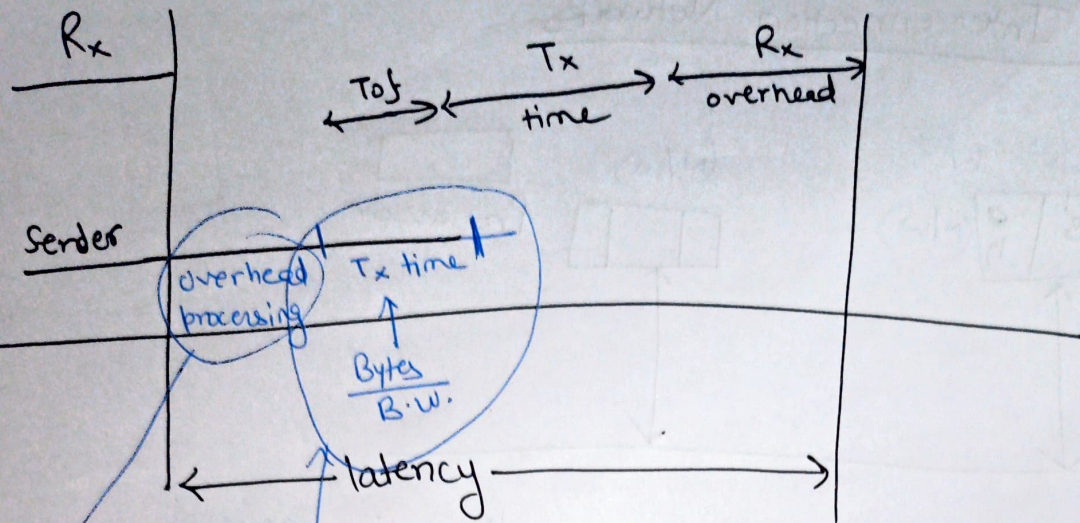
(Important thing)

Overhead

↳ Type of things we add to Data at source end

Packet / message goes out b/w two nodes





Can do better h/w

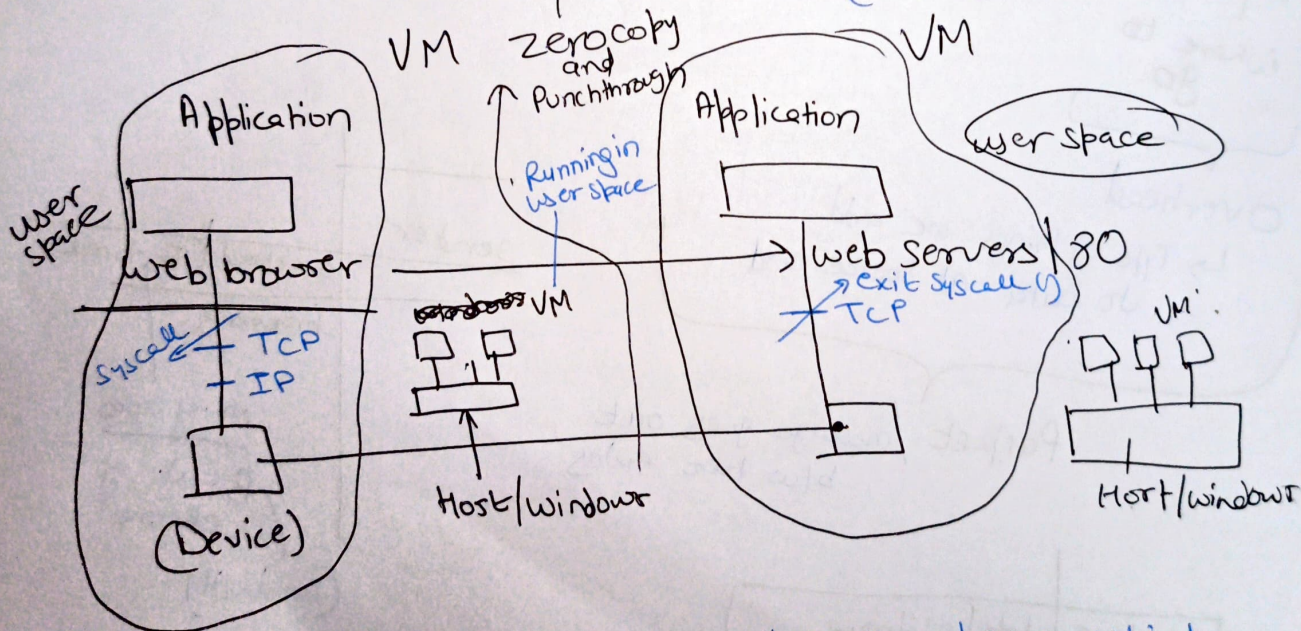
Can't do much here except changing channel

TTL
MTU
Need of these?

effective B.W.?

(OS)
(Learn how it's done)

Have you seen TTL expired? (in system)?



syscall means payload of both sides are been copied to kernel stats
they are bit to crc check (Data layer)
physical layer do encoding

VM-where concept- You can Run a unmodified system on Host
(without changing kernel)

~~these~~

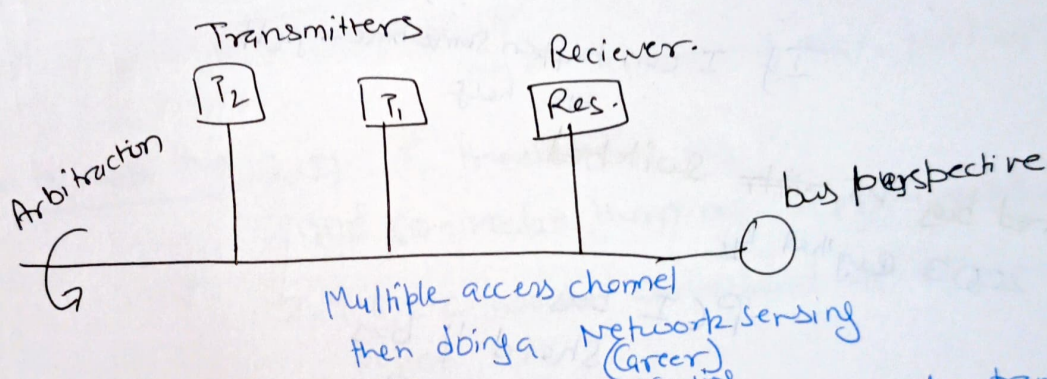
Why zero copy or punchthrough

↳ Running VM on windows was giving extra overhead
and more problems so to have single data/overhead

To study N/w we have to look at

① Topology : Possible paths that a packet can follow

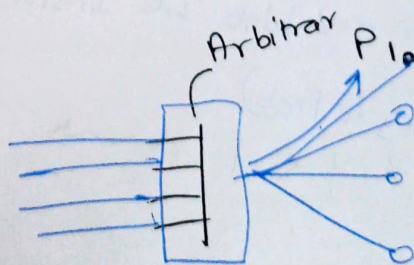
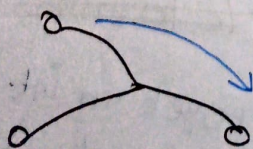
② Routing : If you have 100 path which is allowed



- If noone talking, channel free you are okay to transfer

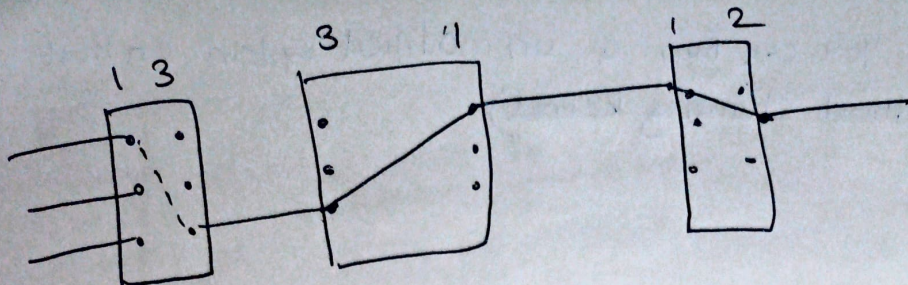
- n users
- 1 winner
- (n-1) losers

③ Arbitration : when are the paths available



Problem
- 1 winner always
- might have starvation
- not fair

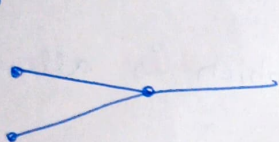
All want to go P₁, Arbitrar
chooses who goes P₁ and queues
rest



mesh Switched Network
 NO sharing, NO carrier sensing, NO packet getting stuck.

④ Switching

How and when are paths allocated
 Any N/w topology that we study has to be checked in these terms



bidirectional line

Info can go both ~~ways~~ Dir at same time

If I can happen same time - full else half

Shared bus vs ~~with~~ switched.

till 2003 ~~evolved by~~

PCI bus

Shared 11 bus

66-133 MHz

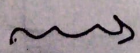
— Inter connection N/w basics (Slide)

Indirect N/w - shared medium N/w

Direct - end points sit Inside

(Everything is Prob)

$$0 < x, y < 1$$



$$0 < \frac{x}{y} < \infty$$

~~precision~~

~~precision~~

→ want to get an int
 (use Round/floor/ceil)

$$0 < (\text{round/floor/ceil}) \left(\frac{x}{y} \right) < \infty$$

(Int)

$$\Pr \left[\text{round} \left(\frac{x}{y} \right) \cdot y \text{ is even} \right] \rightsquigarrow 0.4546 \quad \frac{5-\pi}{5}$$

$$\Pr \left[\text{floor} \left(\frac{x}{y} \right) \text{ is even} \right] \rightsquigarrow 0.65 \log_2 2$$

[Integral way $\rightsquigarrow$ complicated]

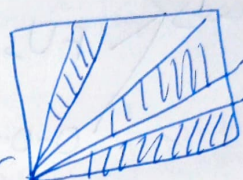
How $\rightarrow$ Calculus / conditional prob etc

Another interesting way $\rightsquigarrow$ counting ~~principle~~ principle [Depends on Distributions]

First even $\rightarrow 2$

$$0 < \frac{x}{y} < 2$$

$$x = y/2$$



Unit Square

Taylor expansion of each portion

Reason to ask this : If we treat them as RV
and co-relate them as departure time
2, 4, 6, etc would correspond to sizes of
buffer

Queuing theory of N/w depends on above 2